

Session 2: AI for Coding – Practical Workflows and Demos

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Plan for Today: hands-on workflows

1. Coding with Claude Code overview

- ▶ Writing and debugging code from plain English
- ▶ “Vibe coding”: building a complete tool without writing a line of code

2. Data analysis demos with Stata

- ▶ Stata example with Claude Code from terminal
- ▶ Stata example with Claude Code from VS Code

3. Python in VS Code overview

- ▶ libraries and environments
- ▶ Jupyter extension

4. Data analysis demo with Python

- ▶ Python example with Claude Code from VS Code

Writing and Debugging Code with AI

Writing Code from Plain English

- ▶ You describe the task; AI writes the code — in any language (Stata, R, Python, Julia)
- ▶ This works even if you do not know the language
- ▶ Key skill: writing a good prompt
 - Describe inputs and outputs concretely: “I have a panel dataset with vars X, Y, Z...”
 - Specify what you want verified: “run it and show me the first 5 rows of output”
 - Be explicit about conventions: “use ‘i.year’ for year fixed effects in Stata”
- ▶ ▷ *Placeholder: side-by-side example of prompt → code*

Debugging Code with AI

- ▶ Paste the error message and ask: “what is wrong and how do I fix it?”
- ▶ With Claude Code in agentic mode: it sees the error itself and iterates automatically
- ▶ Particularly powerful for:
 - Obscure Stata/R errors with cryptic messages
 - Merge problems (wrong key, many-to-many, unmatched observations)
 - Output that looks right but is subtly wrong (off-by-one, wrong unit of observation)
- ▶ Important caveat: AI can fix the error while introducing a different bug
 - Always verify output against known benchmarks or summary statistics

“Vibe Coding”: Building Tools Without Writing Code

- ▶ Concept: describe what you want at a high level; AI writes, runs, and refines all the code
- ▶ You act as the product manager, not the programmer
- ▶ Examples relevant to economists:
 - “Build a script that downloads monthly CPI from FRED, merges with my wage data, and plots real wages by year”
 - “Create a do-file that runs my DiD specification with all the standard robustness checks and exports tables to LaTeX”
 - “Write a web scraper that pulls contract award data from USASpending.gov”
- ▶ When it works well; when to be cautious

Automating Repetitive Data Tasks

- ▶ Common tasks AI handles easily:
 - Cleaning and reshaping messy data (wide ↔ long, renaming, recoding)
 - Labeling variables and values from a codebook
 - Merging multiple datasets on complex keys
 - Generating a full set of summary statistics or balance tables
- ▶ Practical workflow for cleaning:
 1. Show Claude the raw data structure (variable list, codebook, first rows)
 2. Describe the cleaning steps in plain English
 3. Review the output, ask for changes, iterate
- ▶ ▷ *Placeholder: example or screenshot*

Working with Claude Code: Ask, Plan, Auto

Three Modes of Operation

▶ **Ask before edits** (safest):

- Claude proposes each file edit before making it; you approve or reject
- Best for: getting started, when you want full control, unfamiliar codebases

▶ **Plan mode** (recommended for most tasks):

- Claude writes a full plan in Markdown; you review and approve before anything runs
- Produces clean, commented code with an explicit audit trail
- Can view and revisit session history

▶ **Auto mode** (most powerful, most risk):

- Claude acts autonomously until the task is done
- Use for well-defined tasks; always check the output

Managing Your Session: Key Commands

- ▶ `/compact <hint>` — compress context before it overflows; hint steers what Claude keeps
- ▶ `/clear` — fresh start, zero accumulated context; you choose what to carry forward
- ▶ **Plan mode** before long tasks — Claude writes a plan first; avoids wasted runs
- ▶ **CLAUDE.md** — store standing project instructions so Claude knows your setup from the start
- ▶ Scope to specific files — “look only at `clean.do`” cuts tokens and sharpens focus

Demo: Writing a Data Cleaning Script (Ask Mode)

- ▶ ▷ *Demo slide*
- ▶ Task: take a raw dataset, produce a clean analysis file
- ▶ Steps we will walk through:
 1. Point Claude Code at the project folder
 2. Write a minimum working example in “Ask before edits” mode
 3. Claude proposes each change; we approve
 4. Verify output matches expectations
- ▶ What to watch for:
 - Does the code run without errors?
 - Do summary statistics match what we expect?
 - Any implicit assumptions we need to check?

Demo: Cleaning Script Continued (Plan Mode)

- ▶ ▷ *Demo slide*
- ▶ Now run the same task in “Plan mode”:
 1. Ask Claude to write a plan before touching any files
 2. Review the plan in Markdown; request changes to the approach
 3. Approve; Claude executes and produces a clean, commented script
 4. Compare output to Ask-mode version: same result, better code
- ▶ Reviewing session history: how to audit what Claude did
- ▶ Ask Claude to update README with what was done (good practice)

When to Trust AI Code – and How to Verify It

- ▶ AI code can look completely correct and be wrong in subtle ways
- ▶ Verification checklist:
 - Does the code run without errors? (necessary, not sufficient)
 - Do observation counts make sense at each merge/collapse step?
 - Do means and distributions match codebook or prior literature?
 - Can you reproduce a known result (e.g., a published Table 1)?
 - Ask Claude: “what assumptions did you make?” – it will often reveal them
- ▶ General rule: trust AI code for *structure*; verify AI code for *content*

Project Structure and Replication

Project Folder Structure – Best Practices

- ▶ Recommended structure for a research project:
 - /raw_data – original data, never modified
 - /data – cleaned/analysis data (Git-ignored if sensitive)
 - /code – all scripts, tracked by Git
 - /output – tables and figures (regenerated, not tracked)
 - /paper – LaTeX files
- ▶ Handling secure/restricted data (GW Box, FSRDC):
 - Raw data lives on Box or secure server; code lives on Git
 - .gitignore ensures data never enters the repo
 - README documents where raw data comes from and access requirements

CLAUDE.md: Persistent Instructions for Claude Code

- ▶ CLAUDE.md is a plain-text file in your project folder that Claude reads at the start of every session
- ▶ Use it to store standing instructions so you do not repeat yourself:
 - Language and style preferences (“always use Stata 17 syntax”, “comment every merge”)
 - Data structure (“the unit of observation is person-year; key vars are `id` and `year`”)
 - File paths (“raw data is in `/Box/Project/raw/`”)
 - Rules (“never modify files in `/raw_data`”)
- ▶ Claude Code “Skills”: project-level or global instruction sets — similar concept
- ▶ ▸ *Demo: write a CLAUDE.md live*

Building a Replication Package with Claude Code

- ▶ Claude Code can read your entire project and produce a replication package:
 1. Point Claude at the project directory
 2. Ask it to trace the data pipeline from raw files to final tables
 3. Ask it to write a master script that runs everything in order
 4. Ask it to write a README documenting data sources, software, and steps
- ▶ This task would normally take days — Claude does it in minutes
- ▶ You still need to verify: does running the master script reproduce the tables?
- ▶ AEA replication standards: what the package needs to include

Git and Coauthor Workflows in VS Code

Using Git in VS Code

- ▶ ▸ *Demo slide – Tanner*
- ▶ The Source Control panel: see all changed files, stage, commit, push
- ▶ Typical session workflow:
 1. Pull latest version from GitHub at start of session
 2. Write code; Claude Code helps
 3. Stage changed files; write a commit message (or ask Claude to write it)
 4. Push to GitHub at end of session
- ▶ Branches for robustness checks:
 - Create a branch for “alternative_sample”
 - Run the checks; if they hold, merge back into main

Working with Coauthors and RAs

- ▶ All coauthors clone the same GitHub repo — everyone has the same code
- ▶ RA workflow:
 - RA works on a branch; submits a “pull request” when done
 - You review the changes before they enter the main code
 - If something breaks, you can always roll back
- ▶ Common pitfalls:
 - Merge conflicts: two people edit the same file — Git flags the conflict; Claude Code can resolve it
 - Committing data files accidentally — prevent with `.gitignore`
- ▶ Overleaf integration: sync your LaTeX repo so the paper drafts with the code

Python and Jupyter in VS Code

Python in VS Code: Environments

- ▶ ▷ *Demo slide – Tanner*
- ▶ Anaconda manages Python installations and packages
- ▶ Each project gets its own **environment**: a fixed set of packages and versions
 - Prevents conflicts between projects
 - Ensures others can reproduce your results exactly
- ▶ Ask Claude Code to create and activate an environment for you

Running Python Code with Jupyter

- ▶ ▷ *Demo slide – Tanner*
- ▶ Three modes in VS Code:
 - **Jupyter notebook** (.ipynb): interactive cells, inline output – good for exploration
 - **Interactive window**: run .py scripts cell-by-cell – best of both worlds
 - **Script** (.py): run the full pipeline – best for production code
- ▶ Claude Code can write, run, and debug Python in any of these modes

Python Setup in VS Code

- ▶ ▷ *Demo slide – Tanner*
- ▶ Required extensions: Python, Jupyter
- ▶ Managing environments:
 - Each project gets its own environment (set of packages + versions)
 - Prevents conflicts between projects
 - Ask Claude Code to create and activate an environment for you
- ▶ Running code:
 - **Jupyter notebook** (`.ipynb`): interactive cells, inline output – good for exploration
 - **Jupyter interactive window**: run `.py` scripts cell-by-cell – best of both worlds
 - **Script** (`.py`): run the full pipeline – best for production code

Python for Economists: Common Tasks

- ▶ *Placeholder: example or screenshot*
- ▶ Data analysis with `pandas`: merging, collapsing, reshaping — analogous to Stata
- ▶ Econometrics: `statsmodels`, `linearmodels` (panel data, IV)
- ▶ Data collection: `requests`, `BeautifulSoup` for scraping (Session 3)
- ▶ Figures: `matplotlib`, `seaborn` — publication-quality plots
- ▶ Claude Code can translate your Stata code to Python (or vice versa) — useful for:
 - Running analyses you cannot do in Stata
 - Working with large datasets where Python is faster

Stata with Claude Code

Using Stata with Claude Code

- ▶ ▸ *Demo slide – Elira*
- ▶ Claude Code works with Stata via the terminal or VS Code (Stata Enhanced extension)
- ▶ What you can ask Claude to do:
 - Write a do-file from a plain-English description of the analysis
 - Debug a Stata error (paste the error; Claude explains and fixes it)
 - Translate a complex cleaning routine into clean, commented Stata code
 - Export regression tables to LaTeX (esttab, outreg2)
- ▶ Example: [PLACEHOLDER – data analysis example to demonstrate]

Stata Demo: Data Analysis with Claude Code

- ▶ ▷ *Demo slide*
- ▶ Task: *[Placeholder: insert specific empirical example]*
- ▶ Walk-through:
 1. Describe the dataset and goal to Claude in plain English
 2. Claude writes the do-file; we run it
 3. Inspect output, ask for revisions
 4. Ask Claude to add comments and export a summary table

Wrapping Up

Key Takeaways from Sessions 1 & 2

1. AI handles the coding mechanics — your job is to verify the output and drive the analysis
2. Use Plan mode for any non-trivial task; it gives you an audit trail and catches mistakes early
3. Git + Claude Code = version-controlled, reproducible research without learning Git commands
4. CLAUDE.md files carry your preferences across sessions so you do not repeat yourself
5. Replication packages are now a hours-long task, not a weeks-long one
6. The critical skill is *verification*: check outputs, not just that the code runs

Things to Try Before the Next Session

- ▶ Set up VS Code with Claude Code, Git, and Python (use the setup guide)
- ▶ Ask Claude to write a small data task in your preferred language — verify the output
- ▶ Initialize a Git repo for one of your current projects
- ▶ Write a CLAUDE.md file for that project
- ▶ Coming up in Session 3: Finding and Collecting Data
 - Identifying datasets, scraping, APIs, parsing PDFs at scale

Resources

- ▶ Anton Korinek (UVA) — *NBER WP 34202* and GenAIforEcon.org
- ▶ Aniket Panjwani — YouTube tutorials on Claude Code for social scientists
- ▶ Paul Goldsmith-Pinkham — practical guide on VS Code + Stata + AI
- ▶ Tyler Ransom — *tyleransom.substack.com*: honest account of daily AI use
- ▶ Setup guide for this course: [PLACEHOLDER — link or shared folder]
- ▶ CLAUDE.md template: [PLACEHOLDER]
- ▶ Replication package example: [PLACEHOLDER]

Questions?

- ▶ Questions on anything from Sessions 1 or 2?
- ▶ What tasks in your own research would you most like to try this on?